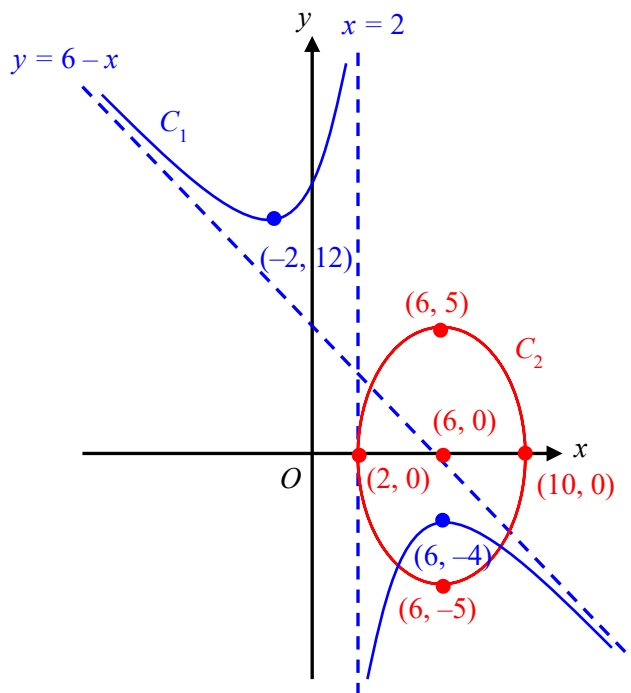


	Solution
1(a)	Differentiating C w.r.t. x, $3x^2 + 3y^2 \frac{dy}{dx} - 3x \frac{dy}{dx} - 3y = 0$ $\frac{dy}{dx} = \frac{3y - 3x^2}{3y^2 - 3x} = \frac{y - x^2}{y^2 - x}$ When $y = 0$, $x^3 = k,$ $x = k^{\frac{1}{3}}$ $\frac{dy}{dx} = x = k^{\frac{1}{3}}$ $y = k^{\frac{1}{3}} \left(x - k^{\frac{1}{3}} \right) = k^{\frac{1}{3}} x - k^{\frac{2}{3}}$
(b)	For tangent to be parallel to y-axis, $\frac{dy}{dx}$ must be undefined. Hence, $y^2 - x = 0$ $x = y^2$ Substituting into C and $k = 3$, $y^6 + y^3 - 3y^3 - k = 0$ $y^6 - 2y^3 - 3 = 0$ Factorising, $(y^3 - 3)(y^3 + 1) = 0$ $y^3 = 3 \quad \text{or} \quad y^3 = -1$ $y = 3^{\frac{1}{3}} \quad \text{or} \quad y = -1$

	Solution
2(a)	Let L be the limit of the sequence $u_{n+1} = 2u_n(1 - u_n)$. $L = 2L(1 - L)$ $2L^2 - L = 0$ $L(2L - 1) = 0$ $L = 0 \text{ or } L = \frac{1}{2}$
(b)	When $u_1 = 0$, $u_2 = 0$, $u_3 = 0$, sequence is convergent. When $u_1 = 0.01$, $u_2 = 0.0198$ (exact), $u_3 = 0.0388$ (3 s.f.), sequence is convergent. When $u_1 = -0.01$, $u_2 = -0.0202$ (exact), $u_3 = -0.0412$ (3 s.f.), sequence is not convergent.

Solution

3(a)



(b) x -coordinate of points of intersection are 4.6391, 7.7172 to 5 sf

Equation of ellipse: $y^2 = 25 \left(1 - \frac{(x-6)^2}{16} \right)$

Required volume

$$= \pi \int_{4.6391}^{7.7172} 25 \left(1 - \frac{(x-6)^2}{16} \right) - \left(\frac{x^2 - 8x + 28}{2 - x} \right)^2 dx$$

$$= 58.7 \text{ units}^3$$

	Solution
4(a)	Normal vector of $p = \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix} \times \begin{pmatrix} 5 \\ 0 \\ t \end{pmatrix} = \begin{pmatrix} 5t \\ 10 \\ -25 \end{pmatrix} = 5 \begin{pmatrix} t \\ 2 \\ -5 \end{pmatrix}$ plane $p: \mathbf{r} \cdot \begin{pmatrix} t \\ 2 \\ -5 \end{pmatrix} = 0$ Since l_1 and p are parallel, $\begin{pmatrix} t \\ 3-t^2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} t \\ 2 \\ -5 \end{pmatrix} = 0$ $t^2 + 6 - 2t^2 - 5 = 0$ $t^2 = 1$ $t = 1 \quad \text{or} \quad t = -1$ Since point A is not on p, $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} t \\ 2 \\ -5 \end{pmatrix} \neq 0$ $t + 4 - 5 \neq 0$ $t \neq 1$ $\therefore t = -1$ (Shown)
(b)	Let θ be the acute angle between l_1 and l_2 $\theta = \cos^{-1} \left \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right $ $= \cos^{-1} \left \frac{4}{6} \right $ $= 0.841 \quad \text{or} \quad 48.2^\circ$
(c)	$\mathbf{a} \cdot \mathbf{n}$ is the perpendicular distance from A to p $ \mathbf{a} \cdot \mathbf{n} = \left \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{30}} \begin{pmatrix} -1 \\ 2 \\ -5 \end{pmatrix} \right $ $= \left \frac{-2}{\sqrt{30}} \right $ $= \frac{2}{\sqrt{30}}$

(d) Let N be the foot of perpendicular from A to p

$$\overrightarrow{ON} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - s \begin{pmatrix} -1 \\ 2 \\ -5 \end{pmatrix} = \begin{pmatrix} 1+s \\ 2-2s \\ 1+5s \end{pmatrix}$$

Since N is on p ,

$$\begin{pmatrix} 1+s \\ 2-2s \\ 1+5s \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ -5 \end{pmatrix} = 0$$

$$-1-s+4-4s-5-25s=0$$

$$s = -\frac{1}{15}$$

$$\therefore \overrightarrow{ON} = \frac{1}{15} \begin{pmatrix} 14 \\ 32 \\ 10 \end{pmatrix}$$

Let B be the point of reflection of A in p

$$\overrightarrow{OB} = 2\overrightarrow{ON} - \overrightarrow{OA}$$

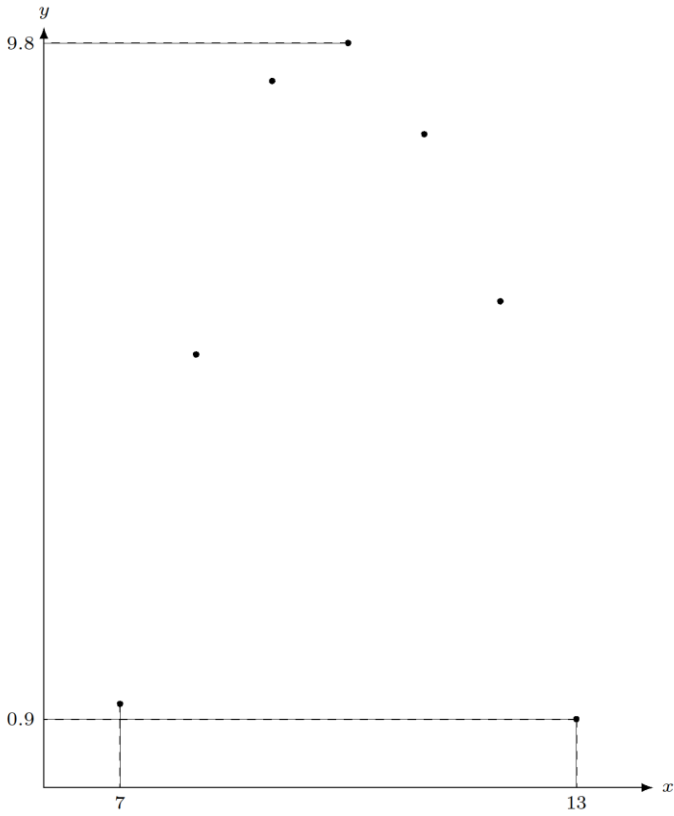
$$= \frac{2}{15} \begin{pmatrix} 14 \\ 32 \\ 10 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$= \frac{1}{15} \begin{pmatrix} 13 \\ 34 \\ 5 \end{pmatrix}$$

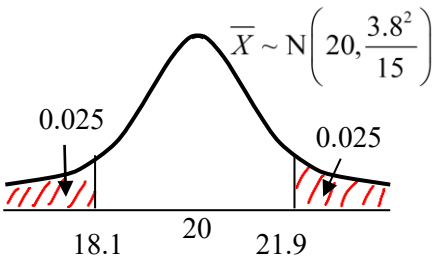
$$\text{line of reflection of } l_2 \text{ in } p: \mathbf{r} = k \begin{pmatrix} 13 \\ 34 \\ 5 \end{pmatrix}, k \in \mathbb{R}$$

	Solution
5(a)	Number of ways to choose the 6 glasses = total number of ways – number of ways with 1 missing colour $= \binom{8}{6} - \binom{4}{1} = 24$ Required number of ways $= 24 \times (6-1)! = 2880$
(b)	Number of ways = $4 \times 4 = 16$
(c)	Let A be the event that the blue glasses are adjacent. Let B be the event that the yellow glasses are adjacent. Then $n(A') = 4! \times \binom{5}{2} \times 2! = 480$ [Explanation: arrange the four non-blue glasses ($4!$), then slot in the blue glasses (${}^5C_2 \times 2!$)] Also $n(A' \cap B) = 3! \times 2! \times {}^4C_2 \times 2! = 144$ [Explanation: group the two yellow glasses ($2!$), arrange this group with the red and green glass ($3!$), then slot in the blue glasses (${}^4C_2 \times 2!$)] So the required number of ways is $n(A') - n(A' \cap B) = 336$.

	Solution
6(a)	Stage 1 Stage 2 Stage 3 Start → Win (p) → Win (0.5p) → Win (0.25p) / Lose (1-0.25p) Start → Win (p) → Lose (1-0.5p) → Win (0.5p) / Lose (1-0.5p) Start → Lose (1-p) → Win (p) → Win (0.5p) / Lose (1-0.5p) Start → Lose (1-p) → Lose (1-p) → Lose (1-0.5p) / Lose (1-0.5p)
(b)	$p(0.5p)(0.5^2 p) > 0.1$ $p^3 > 0.8$ $p > 0.928$
(c)	$P(\text{wins exactly 2 stages out of 3} \mid \text{proceeds to Stage 3})$ $= \frac{P(\text{wins exactly 2 stages out of 3} \cap \text{proceeds to Stage 3})}{P(\text{proceeds to Stage 3})}$ $= \frac{0.9(0.45)(1-0.225) + 0.9(0.55)(0.45) + 0.1(0.9)(0.45)}{1-(0.1)(0.1)}$ $= \frac{0.577125}{0.99} = 0.583$

	Solution
7(a)	$r = 0.00208$
(b)	 There is a non-linear (curved) relationship between x and y.
(c)	$r = -0.997$. This means there is a (very) strong negative linear correlation between y and $(x-10)^2$.
(d)	Equation is $y = -0.987(x-10)^2 + 9.92$ When $x = 17$, $y = -38.4$ This estimate is not valid because $x = 17$ lies outside the valid range of $7 \leq x \leq 13$

	Solution				
8(a)	$E(X - 3) = E(X) - 3 = 4 - 3 = 1$				
	$\text{Var}(X - 3) = E((X - 3)^2) - [E(X - 3)]^2 = 40 - 1 = 39$				
	$\text{Var}(X) = \text{Var}(X - 3) = 39$				
	Alternatively,				
	$E((X - 3)^2) = E(X^2 - 6X + 9) = 40$				
	$E(X^2) - 6E(X) + 9 = 40$				
	$E(X^2) = 40 + 6 \times 4 - 9 = 55$				
	$\text{Var}(X) = E(X^2) - [E(X)]^2 = 55 - 16 = 39$				
(b)	y	0	1	2	3
	$f(y)$	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$
	$P(Y = y)$	$\left(\frac{2}{3}\right)^3$	${}^3C_1\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)$	${}^3C_2\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^2$	$\left(\frac{1}{3}\right)^3$
	$f(y) \times P(Y = y)$	$\frac{8}{27}$	$\frac{2}{9}$	$\frac{2}{27}$	$\frac{1}{108}$
	$E[f(Y)] = \sum_y [f(y) \times P(Y = y)]$ $= \frac{8}{27} + \frac{2}{9} + \frac{2}{27} + \frac{1}{108}$ $= \frac{65}{108}$ (or 0.602 to 3 s.f.)				
(c)	Since sum of probabilities is 1, $p + q + 0.15 + r = 1$ $p + q + r = 0.85$ (1)				
	From $E(W) = -0.2$, $-3p - q + 0.15 + 2r = -0.2$ $-3p - q + 2r = -0.35$ (2) From $E(W^3) = -3.2$, $-27p - q + 0.15 + 8r = -3.2$ $-27p - q + 8r = -3.35$ (3) From GC, solving simultaneous linear equations, $p = 0.2, q = 0.35, r = 0.3$				

	Solution
9(a)	The probability that any student is selected is the same, and the selection of any student is independent of the selection of other students.
(b)	Let X be the queue time at the canteen during peak hours of a randomly chosen EJC student and let μ be the population mean. $H_0 : \mu = 20, \quad H_1 : \mu \neq 20$ Assume that the queue times are normally distributed. Under H_0, $\bar{X} \sim N\left(20, \frac{3.8^2}{15}\right)$ From GC,  Given that H_0 is not rejected at 5% level of significance, $18.1 < \bar{x} < 21.9$
(c)	$\bar{t} = \frac{\sum(t-20)}{50} + 20 = 19.3$ $s^2 = \frac{1}{n-1} \left(\sum(t-20)^2 - \frac{(\sum(t-20))^2}{n} \right)$ $= \frac{1}{49} \left(489 - \frac{(-35)^2}{50} \right)$ $= \frac{929}{98} \text{ or } 9.4796 \approx 9.48 \text{ (3 s.f.)}$
(d)	Let μ be the population mean (of T). $H_0 : \mu = 20, \quad H_1 : \mu < 20$ Under H_0, since sample size $n=50$ is large, by Central Limit Theorem, $\bar{T} \sim N\left(20, \frac{9.4796}{50}\right)$ approximately. At the 10% level of significance, the critical region is $\bar{t} < 19.442$. From data, $\bar{t} = 19.3 < 19.442$ Hence, we reject H_0 and conclude at 10% level of significance that there is sufficient evidence that EuOrder has reduced the queue time at the canteen.

	Solution						
10(a)	Let X be the mass of a randomly chosen apple, in grams. Then $X \sim N(100, 6^2)$. $P(X \leq 105) = 0.79767 = 0.798$ (3 s.f.)						
(b)	Let Y be the mass of a prepared apple, in grams. $E(Y) = 0.7 \times 100 = 70$ $\text{Var}(Y) = 0.7^2 \times 6^2 = 17.64$ Then, $Y \sim N(70, 17.64)$. $P(Y > 78) = 0.028405 = 0.0284$ (3 s.f.)						
(c)	Let T be the total mass of 2 prepared apples, 1 peeled banana, 12 blackcurrants and milk, in grams. $E(T) = 2(70) + 130 + 12(2.5) + 50 = 350$ $\text{Var}(T) = 2(17.64) + 8^2 + 12(0.6^2) = 103.6$ $T \sim N(350, 103.6)$ $P(T \geq m) = 0.9$ Using GC, $m = 336.96 = 337$ (nearest gram)						
(d)	$P(330 < T < 360) = 0.81236$ Let R be the number of Signature smoothies, out of 25, with masses between 330 and 360 grams. Then, $R \sim B(25, 0.81236)$ $P(R \geq 20) = 1 - P(R \leq 19) = 0.677$ (3 s.f.)						
(e)	Let W be the total mass of n blackcurrants and milk, in grams. $E(W) = n(2.5) + 175 = 2.5n + 175$ $\text{Var}(W) = n(0.6^2) = 0.36n$ $W \sim N(2.5n + 175, 0.36n)$ $P(W > 300) \geq 0.95$ Using GC, <table border="1"> <thead> <tr> <th>n</th><th>$P(W > 300)$</th></tr> </thead> <tbody> <tr> <td>52</td><td>0.8761</td></tr> <tr> <td>53</td><td>0.957</td></tr> </tbody> </table> Hence, the least value of $n = 53$.	n	$P(W > 300)$	52	0.8761	53	0.957
n	$P(W > 300)$						
52	0.8761						
53	0.957						